Is Acceleration Quadratic - or Is It Depth Uncertainty?

Geometry, clocks, observer motion and tests of apparent acceleration

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Abstract

Quadratic dependence occurs in several different parts of motion measurement. Under constant acceleration, position is quadratic in elapsed time. In small-angle triangulation with fixed baseline and angular uncertainty, the standard uncertainty of inferred depth grows quadratically with distance. These statements concern different quantities and do not imply one another. This paper derives exact symmetric stereo geometry, its local covariance propagation and the limitations of reciprocal parallax inversion. It gives an explicit example in which a constant parallax zero-point error turns uniform radial motion into apparent acceleration. Observer oscillation, perspective, clock-rate variation and numerical differentiation provide additional sources of curvature in measured trajectories. A joint observation model and controlled changes of baseline, time span and measurement channel are proposed to distinguish these effects. Reproducible synthetic calculations illustrate both false acceleration and recovery of an injected acceleration. No real astronomical acceleration is reclassified by these examples. The result is a mathematical test framework, not a claim that acceleration in general is a depth-estimation artifact.

Central distinction. A standard uncertainty is a measure of incomplete knowledge; a bias is a systematic displacement of an estimator; a physical acceleration is a second derivative of position in a specified dynamical frame and time coordinate. They may interact in an experiment, but are not interchangeable.

Scope. The paper follows the observer-aware research direction of *all at once - maybe?* [1]. The derivations below are self-contained. They concern local Euclidean triangulation and specified clock and motion models. Cosmological distance inference needs a separate forward model and is not replaced by the stereo formulas used here.

Main questions. Which variable is being squared? Does the measured mean trajectory curve, or does only its uncertainty grow? Which independent measurements can separate dynamics from calibration and observer motion?

1 What is actually quadratic?

For one Cartesian component in a fixed inertial frame, let $x(t)$ be position, $v = \mathrm{d}x/\,\mathrm{d}t$ velocity and $a = \mathrm{d}^2x/\,\mathrm{d}t^2$ acceleration. If a is constant,

$$x(t) = x_0 + v_0(t - t_0) + \tfrac{1}{2}a(t - t_0)^2. \tag{1}$$

Position is quadratic in time; velocity is linear; acceleration is constant. A force law may instead specify a different dependence, such as an inverse-square gravitational acceleration as a function of radius. There is no universal statement that "acceleration is quadratic" without naming the variable and physical model.

Quantity	Example relation	Meaning
Position	$x - x_0 - v_0\Delta t = \frac{1}{2}a\Delta t^2$	Constant physical acceleration
Depth standard uncertainty	$\sigma_r \simeq r^2\sigma_\omega / B$	Local sensitivity of triangulation
Depth variance	$\sigma_r^2 \simeq r^4\sigma_\omega^2 / B^2$	Square of the preceding uncertainty
Transverse uncertainty	$\sigma_\perp \simeq r\sigma_\theta$	Position uncertainty from angular uncertainty

Acceleration has units $\mathrm{m\,s^{-2}}$; depth uncertainty has units m. Matching a visible quadratic shape does not identify the underlying quantity.

1.1 Perspective curvature without a force

Even a noise-free observed coordinate can have nonzero second derivative when the Cartesian motion is uniform. For a fixed inertial observer, write $r = |x|$, $n = x/r$, $v_r = n \cdot v$ and $v_\perp = (I - nn^T)v$. If $\ddot{x} = 0$,

$$\dot{r} = v_r, \quad \ddot{r} = \frac{|v_\perp|^2}{r}, \quad \dot{n} = \frac{v_\perp}{r}, \quad (I - nn^T)\ddot{n} = -2\frac{v_r}{r}\dot{n}. \tag{2}$$

For example, differentiation of $r^2 = x \cdot x$ gives $\ddot{r} = (|v|^2 - v_r^2)/r$. Differentiating $\dot{x} = \dot{r}\,n + r\dot{n}$ and projecting perpendicular to n gives the last relation. The full direction acceleration also contains $-|\dot{n}|^2 n$.

Thus a changing range rate or angular proper motion can arise from projection of uniform motion. This is a deterministic geometric effect, distinct from uncertainty. Equations (2) use equal coordinate time and a fixed observer; light-time and observer motion require additional modeling.

2 Two viewing directions and the uncertainty of depth

Consider two ideal observation points at $(-B/2, 0)$ and $(B/2, 0)$, observing a target at $(0, r)$. Here $B > 0$ is the known transverse baseline and ϖ is the total angular disparity. Angles in derivatives are in radians. Elementary geometry gives

$$\varpi = 2 \arctan \frac{B}{2r}, \qquad r = \frac{B}{2} \cot \frac{\varpi}{2}. \tag{3}$$

This idealization fixes the baseline orientation, target geometry and observation epoch. Moving targets observed at different times need propagation to a common state before the two angles are combined.

Proposition 2.1 (Exact local sensitivities)**.** *For* (3),

$$\frac{\partial r}{\partial B} = \frac{r}{B}, \qquad \frac{\partial r}{\partial \varpi} = -\frac{B}{4} \csc^2 \frac{\varpi}{2} = -\left(\frac{r^2}{B} + \frac{B}{4} \right). \tag{4}$$

Proof. Differentiate the cotangent expression. Then use $\csc^2 u = 1 + \cot^2 u$ and $\cot(\varpi/2) = 2r/B$. □

Let C be the covariance matrix of the estimated inputs (B, ϖ). First-order propagation is $\sigma_r^2 \simeq J C J^T$, where

$$J = \left(\frac{r}{B}, \; -\frac{r^2}{B} - \frac{B}{4} \right). \tag{5}$$

This local linearization, including input correlations, follows the usual measurement-model approach [2]. It is not an exact distributional identity.

If $r \gg B$ and uncertainty is small compared with the disparity, then

$$r \simeq B/\varpi, \tag{6}$$

$$\sigma_r^2 \simeq \frac{r^2}{B^2} \sigma_B^2 + \frac{r^4}{B^2} \sigma_\varpi^2 - 2 \frac{r^3}{B^2} \operatorname{Cov}(B, \varpi). \tag{7}$$

With negligible baseline uncertainty,

$$\boxed{\sigma_r \simeq \frac{r^2}{B} \sigma_\varpi}, \qquad \frac{\sigma_r}{r} \simeq \frac{\sigma_\varpi}{\varpi}. \tag{8}$$

For example, $B = 10\,\mathrm{m}$ and $\sigma_\varpi = 10^{-6}\,\mathrm{rad}$ give local standard uncertainties of 0.1 m at 1 km and 10 m at 10 km. These are hypothetical inputs, not specifications of a telescope.

Two angular errors also need their correlation: $\sigma_\varpi^2 = \sigma_1^2 + \sigma_2^2 - 2 \operatorname{Cov}(\theta_1, \theta_2)$ when disparity is their difference. Some common errors cancel; differential errors do not.

3 Growing uncertainty does not by itself create acceleration

Suppose position measurements have the correctly specified mean model

$$y_i = r_0 + vt_i + \epsilon_i, \qquad \mathbb{E}[\epsilon_i] = 0, \qquad \mathrm{Var}(\epsilon_i) = \sigma_i^2. \tag{9}$$

The σ_i may increase as the target recedes. The expected trajectory remains linear. Larger scatter produces less certain acceleration estimates; it does not require their expectation to be nonzero. A selected finite realization can nevertheless look curved. Selection effects and an incorrectly specified estimator require separate analysis.

3.1 Reciprocal inversion is a nonlinear operation

The observation $\widehat{\varpi} = \varpi + \epsilon$ is different from an unbiased range observation. For the small-angle inverse and $|\epsilon| \ll |\varpi|$,

$$\frac{B}{\varpi + \epsilon} = \frac{B}{\varpi} - \frac{B}{\varpi^2}\epsilon + \frac{B}{\varpi^3}\epsilon^2 + O(\epsilon^3). \tag{10}$$

For a noise law with suitable bounded support away from the pole, the leading local bias for zero-mean noise is therefore

$$\mathbb{E}[\widehat{r}] - r \simeq \frac{B}{\varpi^3}\sigma_\varpi^2 = \frac{r^3}{B^2}\sigma_\varpi^2. \tag{11}$$

This is an approximation with assumptions on the distribution and the remainder. For an unrestricted Gaussian disparity, the reciprocal has a pole at zero and its ordinary mean and variance need not exist. A high-signal local expansion must not be presented as an exact Gaussian reciprocal expectation.

Distance estimation from noisy parallax is consequently an inference problem, not unconditional inversion of a measured number [4, 5]. A forward likelihood can retain negative measured parallaxes as noisy observations without assigning a negative physical distance. If a prior is used, state and test it. The JCGM nonlinearity amendment likewise calls for higher-order treatment or Monte Carlo propagation when first-order approximations are inadequate [3].

3.2 Why normalizing the scale does not remove missing information

For $r = B/\varpi$, the relative condition number with respect to ϖ is one: $|(\varpi/r)(\partial r/\partial \varpi)| = 1$. Yet for fixed *absolute* angular uncertainty, σ_ϖ/ϖ increases because ϖ gets smaller. Dimensionless coordinates can expose this problem; they cannot supply new angles or eliminate an unknown calibration offset.

4 An explicit false-acceleration mechanism

Consider the declared small-angle observation model

$$\varpi_{\mathrm{measured}}(t) = \frac{B}{r(t)} + b, \qquad r(t) = r_0 + vt, \tag{12}$$

with constant baseline B, constant zero-point error b and constant physical radial velocity v. Suppose an analyst ignores b and inverts the measured disparity.

Proposition 4.1 (A constant angular offset can curve inferred motion)**.** *Let $\kappa = b/B$ and restrict the interval to $r > 0$ and $1 + \kappa r > 0$. Then*

$$\widehat{r} = \frac{r}{1+\kappa r}, \qquad \dot{\widehat{r}} = \frac{v}{(1+\kappa r)^2}, \qquad \ddot{\widehat{r}} = -\frac{2\kappa v^2}{(1+\kappa r)^3}. \tag{13}$$

Proof. Invert (12), differentiate using $\dot{r} = v$, then differentiate again. The assumed denominator excludes the inverse singularity. □

The true acceleration is zero, while the reconstructed trajectory generally has nonzero second derivative. This is a concrete counterexample to the claim that all curvature of an inferred range must be physical acceleration. It does not establish that a particular measured acceleration has this origin. For small $|\kappa r|$, the bias is $\widehat{r} - r \simeq -\kappa r^2$. The synthetic example uses $B = 10\,\mathrm{m}$, $b = 10^{-5}\,\mathrm{rad}$, $r_0 = 1000\,\mathrm{m}$ and $v = 10\,\mathrm{m\,s^{-1}}$. At $t = 50\,\mathrm{s}$, (13) gives approximately $-1.99 \times 10^{-4}\,\mathrm{m\,s^{-2}}$.

4.1 A degeneracy and a way to break it

One disparity measurement at one baseline cannot determine both r and b. For independent Gaussian angular noise with standard deviation σ_ϖ, the known-offset local information about r is

$$\mathcal{I}_{rr} = \frac{B^2}{\sigma_\varpi^2 r^4}. \tag{14}$$

The corresponding regular unbiased-estimation bound reproduces (8). If b is also unknown, the one-configuration information matrix for (r, b) has rank one. Repeating the same configuration reduces random noise but does not separate those two parameters without further information.

For two different baselines viewing the same target state, assume the same offset b and the model $\varpi_j = B_j/r + b$. Then

$$\frac{1}{r} = \frac{\varpi_1 - \varpi_2}{B_1 - B_2}, \qquad b = \frac{B_1\varpi_2 - B_2\varpi_1}{B_1 - B_2}. \tag{15}$$

This cancellation requires a common offset and controlled geometry and timing. Different instrumental offsets or target motion must be estimated separately.

5 Observer motion, phase and sampling

For a small field and one angular component, a local observation model is

$$\theta_i^{\mathrm{obs}}(t) = \theta_i^{\mathrm{source}}(t) + w(t) + \eta_i(t). \tag{16}$$

Here w is a shared angular pointing error. Subtracting two simultaneous source directions cancels the common term under these assumptions. A real rotation over a wide field has a position-dependent projection; translation produces distance-dependent parallax. A single additive shift is not a universal model of observer motion.

5.1 A slow oscillation can resemble a quadratic trend

Let $w(t) = A\cos(\Omega t + \phi)$. Around $t = 0$,

$$w(t) = A\cos\phi - A\Omega\sin\phi\, t - \tfrac{1}{2}A\Omega^2\cos\phi\, t^2 + O(t^3). \tag{17}$$

A short record can therefore be approximately degenerate with an angular offset, angular velocity and angular acceleration. Extending the observation interval, measuring the observer motion independently or deliberately changing the vibration can test that explanation. With continuous noise-free coverage on an open interval, a nonconstant sinusoid cannot be identical to a quadratic polynomial.

The angular quantity $-A\Omega^2\cos\phi$ is not automatically a physical linear acceleration. Mapping it to a trajectory requires a range and a calibrated geometry. Likewise an event count per second is not necessarily a periodic vibration frequency. Uniform sampling can alias an oscillation; exposure averaging and irregular sampling need their own response models.

5.2 Material-dependent response is an explicit nuisance model

For an ideal linear mechanical mode,

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\mathrm{eff}}}{m_{\mathrm{eff}}}}. \tag{18}$$

Density affects effective mass under fixed geometry and mode shape. Stiffness, mounting and temperature also matter. In an ideal harmonic mode, additional energy changes amplitude rather than frequency; nonlinearity can change that relation. Any proposed field-induced motion must specify the force and response function. It cannot be inferred from density alone.

For a scalar mode phase $\Phi = k\cdot x - \omega t + \phi_0$, sampling along $x_o(t)$ gives

$$\omega_{\mathrm{sampled}} = -\frac{\mathrm{d}}{\mathrm{d}t}\Phi(x_o(t), t) = \omega - k\cdot\dot{x}_o(t). \tag{19}$$

This is a component's coordinate-time phase rate. A full detector model also needs its local clock, response, field transformation and integration time. It does not define a universal frequency of the observer or of space.

6 A time anchor is not a clock-rate calibration

Let t be a declared reference time and let a local clock read

$$\tau = f(t), \qquad f \in C^2, \quad f'(t) > 0. \tag{20}$$

Fixing $f(t_0)$ fixes an epoch. It does not determine f' or f''. For a scalar relative position $s(t) = x(t) - o(t)$,

Proposition 6.1 (Acceleration under a time reparameterization)**.**

$$\frac{\mathrm{d}^2 s}{\mathrm{d}\tau^2} = \frac{\ddot{s}}{(f')^2} - \frac{\dot{s} f''}{(f')^3} = \frac{\ddot{x} - \ddot{o}}{(f')^2} - \frac{(\dot{x} - \dot{o}) f''}{(f')^3}. \tag{21}$$

Proof. Apply $\mathrm{d}/\,\mathrm{d}\tau = (1/f')\,\mathrm{d}/\,\mathrm{d}t$ twice. □

An unknown changing clock rate can create a spurious second derivative. A constant rate error rescales a nonzero acceleration but does not turn uniform relative motion into accelerated motion. A consistent change of coordinates and units changes the description, not the physical predictions of a complete model.

For example, let $s(t) = vt$ and $f(t) = t + \frac{1}{2}\gamma t^2$, with $1 + \gamma t > 0$. Then $\mathrm{d}^2 s/\,\mathrm{d}\tau^2 = -v\gamma/(1 + \gamma t)^3$, despite $\ddot{s} = 0$. This is a specified clock-distortion example, not evidence that actual clocks have that distortion.

6.1 Exact time addressing within UNIQUANT

The preceding paper [1] uses a chosen rational unit

$$q = \frac{75}{1012}\,\mathrm{s}, \quad A = \left\lfloor \frac{t - t_0}{q} + \tfrac{1}{2} \right\rfloor, \quad \rho = \frac{t - t_0}{q} - A, \qquad t = t_0 + q(A + \rho). \tag{22}$$

Retaining the remainder retains the model timestamp. Keeping only A introduces an address error at most $q/2$, about 37 ms. Such addresses alone would be unsuitable for high-precision laser timing. This is why exact representation, instrumental time resolution and calibrated physical accuracy must be distinguished. The coordinate unit does not impose a physical jump interval.

An epoch is a specified event and time convention. It does not require an absolutely stationary object. Observer trajectories, clock transformations and synchronization uncertainties must instead be stated in the measurement model.

7 Estimating acceleration with an uncertainty model

For observations in a chosen one-dimensional range coordinate, write

$$y = X\beta + \epsilon, \qquad X_i = (1, t_i, \tfrac{1}{2}t_i^2), \quad \beta = (r_0, v_0, a)^T. \tag{23}$$

If $\mathbb{E}\epsilon = 0$, $\mathrm{Cov}(\epsilon) = \Sigma$ is known and positive definite, X has full column rank and the mean model is correct, generalized least squares gives

$$\widehat{\beta} = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}y, \qquad \mathrm{Cov}(\widehat{\beta}) = (X^T\Sigma^{-1}X)^{-1}. \tag{24}$$

Linearity gives $\mathbb{E}\widehat{\beta} = \beta$. This remains true for heteroscedastic noise. Increasing uncertainty changes the covariance, not this expectation. Gaussian noise is not needed for those identities; it is needed for some exact Gaussian test interpretations. If Σ is estimated from the observations, its uncertainty must also be accounted for.

7.1 Differentiation amplifies errors

For independent equal-variance position errors at three equally spaced times,

$$\widehat{a}_i = \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}, \qquad \mathrm{Var}(\widehat{a}_i) = \frac{6\sigma_r^2}{h^4}. \tag{25}$$

This central difference has a truncation error of order h^2 for sufficiently smooth motion. Shrinking h reduces that deterministic approximation error but amplifies measurement noise. Correlated errors require the full quadratic form in their covariance; overlapping estimates are themselves correlated.

7.2 Fit the original observables when possible

Rather than first inverting noisy angles and then fitting a polynomial, use a forward model for the measured angles, ranges, clocks and environmental channels. A schematic angular model is

$$\varpi_i = \mathcal{P}\big(x(t_i;\beta), o_1(t_i), o_2(t_i), R_1, R_2\big) + b_i + w_i + \eta_i. \tag{26}$$

The projection $\mathcal{P}$ includes the relevant geometry. Retarded emission times, refraction and frequency-dependent responses must be added when relevant. Gaia astrometric processing, for example, explicitly includes spacecraft attitude and instrument calibration [6].

A claimed acceleration must remain distinguishable from allowed nuisance parameters. For a differentiable model, inspect the Jacobian's column rank and singular values after fixing coordinate freedoms. A rank defect reveals an unresolved direction in the linearized model; it does not by itself prove nonlinear nonidentifiability. Full column rank is sufficient for regular local identifiability, while global uniqueness and model correctness require further analysis.

8 Experiments that distinguish the alternatives

The objective is to predict how a candidate effect changes under an intervention. Simply fitting a curved trajectory is insufficient to identify its cause.

Change	Diagnostic prediction	Required control
Increase baseline B	Random angular depth uncertainty decreases approximately as $1/B$	Same angular precision and calibrated geometry
Use two baselines	A shared additive angular offset can be separated by (15)	Same target state and common offset model
Extend time span	A sinusoidal nuisance departs from its local quadratic approximation	Stable timing and adequate cadence
Measure observer motion	Predict the common angular component and its derivatives	Calibrated sensor alignment and response
Add ranging or Doppler	Test one trajectory against multiple observables	Clock, propagation and shared-error covariance
Change material or orientation	Test a specified response function	Temperature, loading, geometry and field conditions

For a stationary reciprocal vacuum path, a basic two-way range relation is

$$r = \frac{c}{2}\left(\Delta\tau - \tau_{\text{instrument}}\right). \tag{27}$$

Actual moving-target measurements require light-time geometry, clocks and propagation corrections. An extra unmodeled timing delay $\delta\tau$ produces range bias $c\delta\tau/2$. A broad beam affects the footprint and return statistics, but does not make range precision identical to transverse image resolution.

8.1 A real example of calibration, not a universal failure claim

The APOLLO lunar-ranging analysis reports a historical median nightly range uncertainty of 1.7 mm and approximately 1 mm accuracy and precision after its independent timing-calibration corrections. These are aggregated range observables. The reported 0.4% calibration error concerned a hardware timing component, not 0.4% of the Earth–Moon distance [7]. This is a concrete example of why calibration and uncertainty budgets matter even when measurements are very precise.

Independent channels can share errors. Their covariance and calibration provenance must be retained rather than assuming that adding instruments automatically removes every common effect. A proposed extra field must predict a specific residual after these controls, with units, coupling and dependence on the intervention specified.

9 Reproducible synthetic diagnostics

The source package includes a deterministic Python programme with fixed random seeds. Its data are simulated; none are observations of stars, planets or a new field. The purpose is to verify the mechanisms and distinguish their statistical meanings.

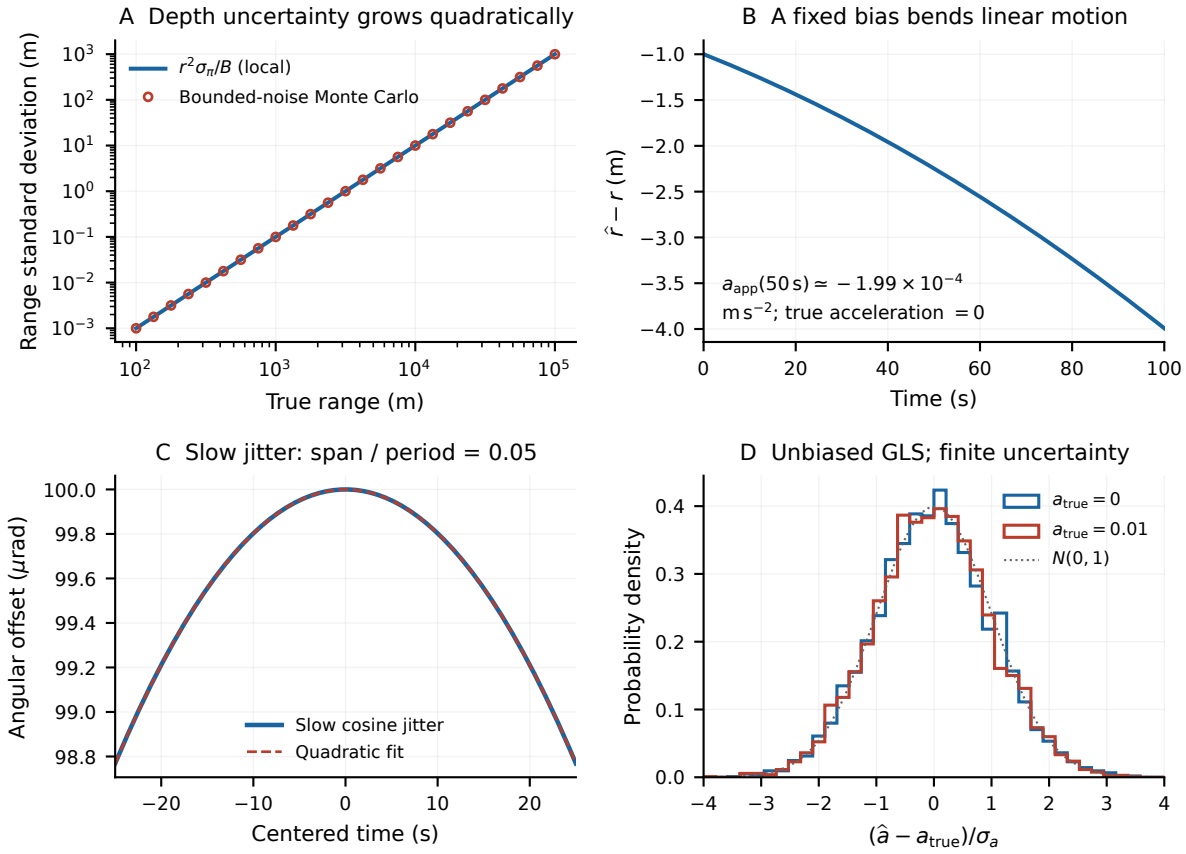


Figure 1: Synthetic examples. A: local depth uncertainty grows approximately as r^2 for fixed baseline and angular noise. B: a constant disparity zero-point error curves the reconstructed range of a uniformly moving target. C: a slow observer oscillation resembles a quadratic over a short window. D: standardized GLS estimation errors for zero and nonzero acceleration injections agree with the propagated uncertainty. Every panel uses the declared finite model, not an astronomical inference.

Diagnostic	Result	Status
Parallax inversion	SD ratio to local law: 0.9964–1.0014	Monte Carlo
Fixed zero point	$a_{app}(50\,s) = -1.99 \times 10^{-4}\,m\,s^{-2}$	Analytic
Slow jitter, $T/P = 0.05$	$\max\lvert w - w_{fit}\rvert = 5.72 \times 10^{-10}$ rad	Numerical
GLS uncertainty	$\sigma_a = 4.18 \times 10^{-5}\,m\,s^{-2}$	Analytic
GLS, injected $a = 0$	$\mathrm{mean}(\hat{a}) = -5.66 \times 10^{-7}\,m\,s^{-2}$	Monte Carlo
GLS, injected $a = 0.01$	$\mathrm{mean}(\hat{a}) = 9.99904 \times 10^{-3}\,m\,s^{-2}$	Monte Carlo

For the inversion experiment, bounded uniform angular noise avoids the reciprocal pole and has a declared variance. The GLS example instead simulates unbiased range observations with known variances; it is not claimed to be identical to nonlinear parallax inversion. The distinction is intentional: the two experiments isolate different mechanisms. Assertions check the analytic identities, sampling statistics and stated synthetic parameters. Numerical success does not establish that one of these mechanisms explains an unspecified real acceleration.

10 Interpretation and a practical decision rule

An uncertain distance, an erroneous reconstructed trajectory and a physical acceleration can coexist. The relevant question is not which word to choose, but which forward model predicts the observed data across controlled configurations.

First identify the observable. Record angles, time intervals, received frequencies or detector positions before derived ranges and accelerations. Keep units, observation epochs and calibration settings.

Then identify the mechanism. Test random angular scatter, angular zero points, perspective, observer movement and clock variation as separately parameterized effects. A noise envelope growing as r^2 is not itself a deterministic force law.

Require an identifying comparison. Vary a baseline or observation window, use known reference targets and clock comparisons, or add a calibrated measurement channel. If acceleration and a nuisance parameter remain degenerate, report that limitation rather than selecting one explanation by preference.

Validate a prediction. Estimate parameters on one configuration and predict held-out times, baselines or instruments. For a proposed material-dependent field response, preregister its sign, amplitude, direction and frequency dependence before examining the comparison data. Arbitrary per-observation corrections make a model unfalsifiable.

10.1 What the derivations establish

They establish a quadratic depth-standard-uncertainty law under stated assumptions, an explicit calibration mechanism for false acceleration and exact formulas for specified perspective and clock effects. They also show why correctly modeled zero-mean scatter need not bias an acceleration estimate.

They do not establish a universal maximum useful distance, a smallest physical time step, a cosmic fundamental frequency or a reinterpretation of all observed accelerations. A claim about cosmic acceleration, in particular, would need the appropriate cosmological observables and a complete competing model.

10.2 Answer to the title question

For constant acceleration, *position* is quadratic in time. Under fixed angular precision and baseline, *depth standard uncertainty* is approximately quadratic in distance. Neither statement substitutes for the other. Some apparent accelerations can be generated by a miscalibrated or incomplete observation model; deciding whether that happened in a particular case requires discriminating data.

Attribution, reproducibility and references

Joern Christian Anders proposed the observer-motion and depth-uncertainty research question developed here. The paper was drafted with AI assistance. The analysis uses standard kinematics, geometry, uncertainty propagation and statistical estimation; no novelty claim is made for those foundations. Its propositions are proved within their declared assumptions. The numerical examples are reproducible illustrations.

The source archive contains `paper.tex`, `reproduce.py`, `results.json`, `results_table.tex`, the vector figure, build instructions and a checksum manifest. Run the Python programme before compiling the paper twice with pdfLaTeX. The bibliography identifies external methodological sources; none of their datasets is redistributed. Online sources were checked on 6 November 2026.

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